

§2.2: Sandwiched Singularities, Extended Graphs, and Decorated Germs

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1. LEMMA V (AS SEEN FROM THE PAPER)

Lemma 1.1. *Up to topological equivalence, there are at most $\text{mult } X$ choices of plane curve germs with smooth branches representing $(X, 0)$.*

Proof. Let G be the fixed minimal normal crossings resolution graph for the rational surface singularity $(X, 0)$. We bound the number of topologically distinct decorated graphs G' by constructing a rigid graph G'' .

Construction of the Rigid Auxiliary Graph G''

Let G' be a valid decorated graph satisfying the conditions of Proposition 2.2. By Proposition 2.4, the number of smooth branches (curvetas) added to the minimal graph G to form G' is:

$$m = \text{mult } X - 1$$

Each of these m branches corresponds to a leaf vertex with self-intersection -1 .

We now construct an auxiliary tree G'' by adding exactly one additional leaf vertex v_{root} with self-intersection -1 attached directly to the root vertex of G . By construction, G'' satisfies the following identity for every vertex $v \in G''$:

$$v \cdot v + d''(v) = 0$$

where $d''(v)$ denotes the valency of the vertex v within G'' .

Counting the (-1) -Vertices via Proposition 2.4

Instead of recomputing the arithmetic genus formula for G'' , we directly leverage the vertex count established in Proposition 2.4:

1. The decorated graph G' contains exactly $m = \text{mult } X - 1$ vertices with self-intersection -1 .
2. The graph G'' is formed by adding exactly 1 more (-1) -vertex (v_{root}) to the existing configuration of G' .

Thus, the total number of vertices in G'' with a self-intersection of -1 is:

$$(\text{mult } X - 1) + 1 = \text{mult } X$$

Because G'' is uniquely determined by the fixed minimal resolution graph G and the root attachment rule, the set of these $\text{mult } X$ vertices is fixed and finite.

Combinatorial Bound on G'

Any valid decorated graph G' representing the singularity configuration can be recovered from the master graph G'' by deleting exactly one of its $\text{mult } X$ vertices that possess a self-intersection of -1 .

Consequently, the number of combinatorially distinct ways to choose the extension G' from G is bounded above by the number of (-1) -vertices available to delete in G'' :

$$\text{Total choices of } G' \leq \text{mult } X$$

Potential symmetries or automorphisms of the underlying graph G may cause different choices to yield isomorphic decorated graphs, making $\text{mult } X$ a strict upper bound. This completes the proof. $\square$

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